

Topic 3: Structure and function of living organisms – Pathogens

Students should:		Guidance <i>iLowerSecondary Curriculum reference</i>
3.1	know the difference between the terms <i>pathogen</i> and <i>parasite</i>	<i>Year 9: Structure and function of living organisms – Pathogens</i>
3.2	understand that, in addition to viruses, some bacteria, some fungi and some protoctists cause disease	for example, flu is caused by a virus; cholera and salmonella are caused by bacteria; athlete's foot is caused by a fungus; malaria is caused by a protoctist recall that viruses are smaller than bacteria viruses have no cell wall, no mitochondria and no nucleus, they cannot live independently, they reproduce inside the cells of other living organisms bacteria are single celled organisms that have a cell wall, cell membrane and cytoplasm but no nucleus fungi cells have cell walls, cell membranes, cytoplasm, mitochondria and a nucleus and vary in size from single cell fungi, for example yeast, to large multicellular organisms, for example mushrooms <i>Year 9: Structure and function of living organisms – Pathogens</i>
3.3	understand that antibiotics are effective against bacteria but not against viruses	<i>Year 9: Structure and function of living organisms – Pathogens</i>
3.4	know the problems associated with overuse of antibiotics in humans and farmed animals	<i>Year 9: Structure and function of living organisms – Pathogens</i>
3.5	understand how vaccination helps to prevent the spread of disease in communities	<i>Year 9: Structure and function of living organisms – Pathogens</i>